

ESP32 Noise Monitoring System – Pin Connections

Components: ESP32, MAX9814 Microphone, OLED SSD1306 (SPI), 3 LEDs, and Buzzer.

1. MAX9814 Sound Sensor

MAX9814 Pin	ESP32 Pin	Description
VDD	3.3V	Power supply
GND	GND	Ground
OUT	GPIO34	Analog sound output
GAIN	GND / Floating	Optional gain setting

2. OLED Display (SPI SSD1306)

OLED Pin	ESP32 Pin	Description
GND	GND	Ground
VCC	3.3V	Power
D0 (CLK)	GPIO18	SPI Clock
D1 (MOSI)	GPIO23	SPI Data
RES	GPIO4	Reset
DC	GPIO2	Data / Command
CS	GPIO5	Chip Select

3. LEDs (use 220Ω resistor)

LED Color	ESP32 Pin	Resistor	Status
Green	GPIO14	220Ω	Quiet
Yellow	GPIO12	220Ω	Average
Red	GPIO13	220Ω	Loud

4. Buzzer

Buzzer Pin	ESP32 Pin	Description
+	GPIO27	Control pin
-	GND	Ground

5. ESP32 Quick Pin Summary

Component	ESP32 Pin
MAX9814 OUT	GPIO34
Green LED	GPIO14
Yellow LED	GPIO12
Red LED	GPIO13
Buzzer	GPIO27
OLED MOSI	GPIO23
OLED CLK	GPIO18
OLED DC	GPIO2
OLED RESET	GPIO4
OLED CS	GPIO5